

Chapter 61: Endocrine Disorders — Revision Notes

Williams Obstetrics 26e, pp. 2834–2887 of the PDF. High-yield numbers are in **bold**. The six tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 6 figures are included from the PDF.

Big picture: Many endocrinopathies are **autoimmune** and may ease in pregnancy only to flare postpartum. Pregnancy changes normal values (TBG, total T4, cortisol, aldosterone, prolactin all ↑; TSH ↓ in the 1st trimester), so **use pregnancy-specific ranges**. Key exam areas: **Graves disease treatment (PTU vs methimazole), thyroid storm, overt vs subclinical hypothyroidism (treat overt only), postpartum thyroiditis, pheochromocytoma vs preeclampsia, prolactinoma surveillance, and Sheehan syndrome**. (Diabetes, the most common endocrinopathy, is covered in its own chapter.)

1. Thyroid Physiology and Pregnancy

- **TBG ↑** → total (bound) T4 and T3 ↑.
- **hCG weakly stimulates the TSH receptor** → during the first **12 wk** (peak hCG) free T4 rises and **TSH falls**; maternal TRH becomes undetectable.
- **TSH does not cross the placenta**. Fetal TRH becomes detectable from midpregnancy (static levels).
- **Maternal T4 crosses to the fetus throughout pregnancy** and is vital for fetal brain development, especially before the fetal thyroid works.
 - Fetal thyroid concentrates iodine and makes hormone **after 12 wk**.
 - **Maternal T4 = 30% of fetal serum T4 at term**.
 - Both high and low maternal thyroid function in the 1st trimester are linked to smaller childhood cortical gray matter.

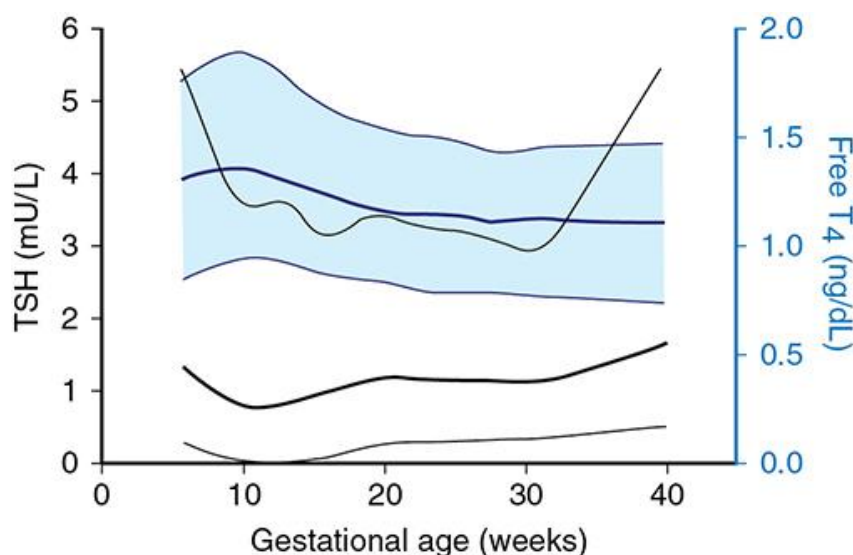


Figure 61-1. Gestational-age-specific TSH (black) and free T4 (blue) in 17,298 women: TSH dips to its nadir near 10–12 wk as hCG peaks; free T4 peaks at the same time then drifts down (median and 2.5th–97.5th percentiles).

Autoimmunity

- Autoantibodies against ~200 thyrocyte components may **stimulate, block or inflame** the gland.
- **Thyroid-stimulating immunoglobulins (TSIs)** activate the TSH receptor → Graves hyperthyroidism and goiter; blocking antibodies may blunt this.

- **Anti-TPO antibodies in 5–15% of gravidas.** Linked in some studies to early pregnancy loss and preterm birth; in >1000 women, **not** preterm birth but **↑** **abruption**. High risk of **postpartum thyroid dysfunction** and lifelong permanent hypothyroidism.

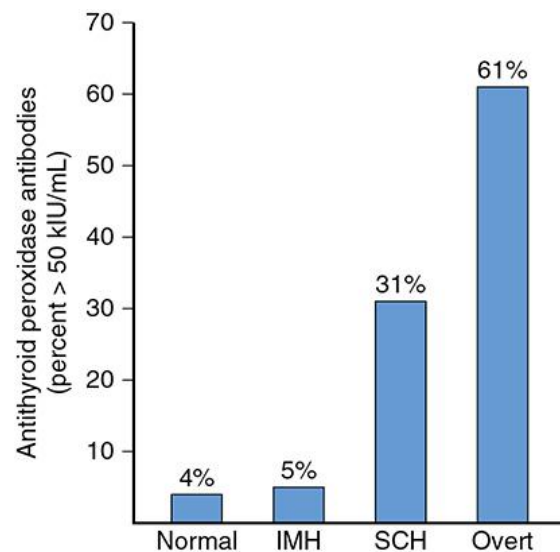


Figure 61-2. Anti-TPO antibody prevalence: normal 4%, isolated hypothyroxinemia 5%, subclinical hypothyroidism 31%, overt hypothyroidism 61%.

Fetal microchimerism

- Fetal cells enter the maternal circulation; fetal lymphocytes can survive **>20 years** and engraft in maternal tissues, including the thyroid.
- **Y-chromosome-positive cells in the thyroid: 60% with Hashimoto, 40% with Graves.** May explain female predominance of autoimmune thyroid disease (and may be protective in some).

2. Hyperthyroidism

Diagnosis

- **Incidence 0.4–1.7%** with gestational-age-appropriate TSH thresholds.
- Clues: **tachycardia beyond normal pregnancy**, thyromegaly, **exophthalmos**, **failure to gain weight despite adequate intake**.
- **Labs: TSH markedly ↓, free T4 ↑.** Rarely T3-toxicosis.

Table 61-1. Incidence of Overt Hyperthyroidism in Pregnancy

Study	Country	Incidence
Wang (2011) ^a	China	1%
Vaidya (2007) ^a	United Kingdom	0.7%
Lazarus (2007) ^b	United Kingdom	1.7%
Casey (2006) ^c	United States	0.4%
Andersen (2016) ^{c, d}	Denmark	0.4–0.7%
Dong (2019) ^e	International	0.64–0.91%

^aScreened in the first trimester. ^bScreened at 9–15 weeks. ^cScreened before 20 weeks. ^dDiagnosed in early versus later pregnancy. ^eSystematic review.

Graves disease in pregnancy

- **Overwhelmingly the most common cause** of thyrotoxicosis in pregnancy; **TSH-receptor antibodies (TRAb)** are specific.
- Course: may **worsen in the 1st trimester (hCG)**, then **improve in the 2nd half** as receptor-antibody titers fall.

Thionamide treatment

	Propylthiouracil (PTU)	Methimazole
Advantages	Partly blocks T4 → T3 ; crosses placenta less	Once-daily; less liver toxicity
Main fetal risk	Rare PTU embryopathy described	Methimazole embryopathy: esophageal or choanal atresia, aplasia cutis (2× major malformations vs PTU/KI in Japanese data)
Main maternal risk	Hepatotoxicity (~0.1%) ; FDA warning 2009; ANCA in ~20% (vasculitis rare)	—
Usual pregnancy dosing	Parkland: 300 or 450 mg/d in 3 doses (occasionally ≥600 mg); nonpregnant start 50–150 mg tid	Nonpregnant: 10–20 mg/d start, then 5–10 mg maintenance

- **ATA: PTU through 16 wk**; no recommendation on switching (both have risks, and the switch may worsen control). **Parkland switches to methimazole after the 1st trimester. Conversion ratio PTU:methimazole = 20:1.**
- **Goal:** lowest dose keeping free T4 **slightly above or at the high-normal range** with TSH still suppressed. **Check free T4 every 4–6 wk.**
- **Side effects of thionamides:**
 - **Transient leukopenia up to 10%** — no need to stop.
 - **Agranulocytosis ~0.3%** — sudden, not dose related → serial WBCs useless; **stop the drug and get a CBC if fever or sore throat.**
 - Hepatotoxicity ~0.1% (serial LFTs don't prevent it); rash **3–5%**.
 - Thionamides may **treat a thyrotoxic fetus** (TRAb crosses the placenta).
- **Subtotal thyroidectomy:** after medical control; for nonadherence or drug toxicity; **best in the 2nd trimester.** Risks: parathyroid removal, recurrent laryngeal nerve injury.
- **Radioactive iodine is contraindicated** (destroys the fetal thyroid). Inadvertent periconceptional exposure has been reported without obvious harm.
 - **Avoid pregnancy for 6 months** after ablation.
 - Lactating breast concentrates iodine → delay ¹³¹I until **3 months after breastfeeding stops.**

Table 61-2. Pregnancy Outcomes in Women with Overt Thyrotoxicosis

Outcome	Treated and Euthyroid ^a (n = 380)	Uncontrolled Thyrotoxicosis ^a (n = 90)
Maternal		
Preeclampsia	40 (10%)	15 (17%)
Heart failure	1	7 (8%)
Death	0	1
Perinatal		
Preterm delivery	51 (16%)	29 (32%)
Growth restriction	37 (11%)	15 (17%)
Stillbirth	0/59	6/33 (18%)
Thyrotoxicosis	1	2
Hypothyroidism	4	0
Goiter	2	0

^aData presented as n (%). From Davis, 1989; Kriplani, 1994; Luewan, 2011; Medici, 2014; Millar, 1994.

- Outcome depends on **metabolic control**; excess T4 → miscarriage, preterm birth. Maternal hyperthyroidism also linked to child epilepsy and autism.

Fetal and neonatal effects

- Most perinates are euthyroid. **Neonatal hyperthyroidism in up to 1%** of infants of women with Graves.
- **Four fetal presentations:**
 1. **Goitrous thyrotoxicosis** — **TShs cross the placenta**; can cause **nonimmune hydrops and fetal death**. Best predictor = maternal **TRAb, especially >3× upper normal**.
 2. **Goitrous hypothyroidism** — from **maternal thionamides** (only 4 of 239 newborns hypothyroid; no long-term IQ effects). Remote if maternal free T4 kept upper-therapeutic/mildly thyrotoxic.
 3. **Nongoitrous hypothyroidism** — transplacental **TSH-receptor blocking antibodies**.
 4. **Fetal thyrotoxicosis after maternal ablation** (surgery or ¹³¹I) — TShs persist and cross.
- Low-risk mothers (no 3rd-trimester antithyroid drugs or no antibodies): no goiters (31 women). Higher-risk (41): **27% fetal goiter at 32 wk** (7 of 11 hypothyroid).
- **ATA: check TRAb in early pregnancy** in women on antithyroid drugs; **repeat at 18–22 wk** if elevated or still on therapy. **ACOG does not recommend this**.
- **Thyrotoxic fetus** → **increase maternal thionamide** even if mother is at target. **Hypothyroid fetus** → **reduce dose ± intraamniotic thyroxine**.
- **Fetal diagnosis:** thyroid volume nomograms exist but routine scans are not recommended. Signs: hydrops, FGR, goiter, tachycardia. **Cord blood sampling only if fetal goiter and fetal thyroid status unclear** (ATA).



Figure 61-3. Term hypothyroid neonate with a large goiter after maternal methimazole 30 mg/d (mother euthyroid at delivery): thionamide-induced goitrous hypothyroidism.

Thyroid storm and thyrotoxic heart failure

- **Thyroid storm** (hypermetabolic state) is **rare**; **thyrotoxic cardiomyopathy with pulmonary hypertension and heart failure is common** in pregnancy — **8%** of uncontrolled thyrotoxicosis.
- Minimal cardiac reserve; decompensation precipitated by **preeclampsia, anemia, sepsis**. Cardiomyopathy is **often reversible**.
- **Management (ICU or L&D special care):**
 1. **Thionamide first** (PTU load)
 2. **Iodide 1–2 h later** (blocks release of T₃/T₄): IV sodium iodide, SSKI or Lugol; **lithium 300 mg q6h if iodine anaphylaxis**
 3. **Dexamethasone 2 mg IV q6h × 4 doses** (blocks peripheral T₄ → T₃)
 4. **β-blocker** for tachycardia (propranolol, labetalol, esmolol) — caution in heart failure
 5. Treat **preeclampsia, infection, anemia aggressively before considering delivery**

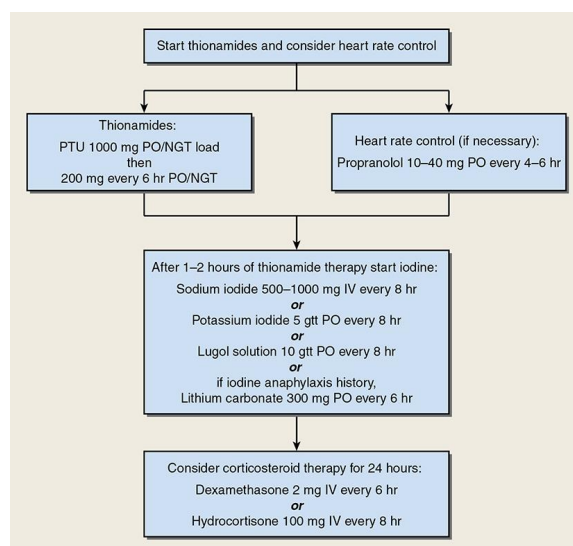


Figure 61-4. Parkland protocol: PTU 1000 mg load then 200 mg q6h ± propranolol 10–40 mg q4–6h → after 1–2 h iodine (NaI 500–1000 mg IV q8h, SSKI 5 gtt q8h or Lugol 10 gtt q8h; lithium if iodine allergy) → dexamethasone 2 mg IV q6h or hydrocortisone 100 mg IV q8h for 24 h.

Gestational transient thyrotoxicosis and hyperemesis

- Biochemical hyperthyroidism in **2–15%** in early pregnancy; common with **hyperemesis** (high T4, low TSH from hCG).
- Antithyroid drugs are NOT warranted.** Values normalize by **midpregnancy**; hCG levels don't correlate with T4/TSH.

Gestational trophoblastic disease

- ↑ **T4 in 25–65%** of molar pregnancies (very high hCG). Early diagnosis → overt hyperthyroidism is less common. **Free T4 normalizes rapidly after evacuation**, parallel with hCG.

Subclinical hyperthyroidism

- Low TSH + normal T4**; prevalence **1.77%** in pregnancy.
- Long term: osteoporosis, cardiovascular morbidity, progression.
- Not associated with adverse pregnancy outcomes** (no link with preeclampsia or GDM).
- Do not treat in pregnancy** (fetal drug effects); surveillance — **~half normalize**.

3. Hypothyroidism

Overt hypothyroidism

- 0.2–1.2%** of pregnancies. Fatigue, constipation, cold intolerance, cramps, weight gain, edema, dry skin, hair loss, **prolonged relaxation phase of reflexes**.
- Overt = ↑ TSH + ↓ T4. Subclinical = ↑ TSH + normal T4.**
- Most common cause: Hashimoto thyroiditis** (anti-TPO); also post-ablative Graves.
- Test** symptomatic women, those with personal/family thyroid history, or **type 1 diabetes**.
- Severe hypothyroidism is uncommon in pregnancy (linked to infertility and miscarriage).

Treatment (ACOG)

- Levothyroxine 1–2 µg/kg/d (~100 µg/d)**; more if athyreotic (thyroidectomy, radioiodine).
- TSH every 4 wk in the first half and at least once in the 3rd trimester**; adjust by **25–50 µg** to a **TSH ~2.5 mU/L**.
- Dose requirement ↑ in ~⅓** (estrogen), starting as early as **5 wk** — especially with no thyroid reserve or ART.
- An anticipatory **25% dose increase** at confirmation of pregnancy is equivalent to TSH-guided adjustment in an RCT (but more TSH suppression). Parkland: TSH at first visit and **each trimester**.

Table 61-3. Pregnancy Complications in 440 Women with Hypothyroidism

Complications (%)	Overt (n = 112)	Subclinical (n = 328)
Preeclampsia	32	8
Placental abruption	8	1
Cardiac dysfunction	3	2
Birthweight <2000 g ^{a,b}	33	32
Stillbirths ^c	9	3 ^c

^aPreterm or term deliveries were the only outcomes reported by Abalovich, 2002. ^bLow birthweight and stillbirth were outcomes reported by Su, 2011. ^cOne infant died from syphilis. Data from Abalovich, 2002; Davis, 1988; Leung, 1993; Männistö, 2009; Su, 2011.

- Untreated overt disease → ↑ adverse outcomes and preterm birth; **adequate replacement minimizes risk**.

Fetal and neonatal effects

- Iodine deficiency causes both maternal and fetal hypothyroidism.
- TSH-receptor **blocking** antibodies cross the placenta. Neonates of mothers with autoimmune thyroiditis: 28% had ↑ TSH on day 3–4, but antibodies were gone by 6 months; **maternal TPO antibodies have no clinically relevant effect** on offspring thyroid function.
- **Fetal hypothyroidism with maternal Hashimoto ≈ 1 in 180,000.**

4. Subclinical Hypothyroidism and Related States

Subclinical hypothyroidism

- **2.3%** of >25,000 women screened in the first half of pregnancy (3.3% in the MFMU screen of >97,000).
- Progression depends on TSH, age, diabetes, antibodies. Nonpregnant: ⅓ normalize; **TSH 10–15 → 19/100 patient-years overt; TSH <10 → 2/100 patient-years**. Nearly all who become overt within 5 yr started with **TSH >10**.
- **Progression during pregnancy is remote**; 17% developed thyroid disease over 20 years (mostly antibody-positive).
- **Associations (observational)**: ↑ preterm birth, abruption, NICU admission (Casey), **~2× severe preeclampsia**, GDM (↑ with rising TSH), diabetes and stillbirth — not seen in all studies.
- **Treatment trials — no benefit**:
 - **CATS**: treating subclinical hypothyroidism/hypothyroxinemia did not improve child IQ at 3 or 9 years.
 - **MFMU RCT (677 women)**: levothyroxine vs placebo — **no difference in pregnancy outcomes or IQ at 5 years** (Table 61-4); no improvement in maternal depression.
- **⚠ Screen only high-risk women** (ACOG, Endocrine Society, ATA, AACE). **No universal screening.**

Table 61-4. Maternal-Fetal Medicine Units Network: Pregnancy Outcomes According to Diagnosis and Treatment of Thyroid Disorders^a

Outcome	Subclinical Hypothyroidism — Thyroxine	Subclinical Hypothyroidism — Placebo	Isolated Hypothyroxinemia — Thyroxine	Isolated Hypothyroxinemia — Placebo
Maternal				
EGA at delivery (weeks)	39.1 ± 2.5	38.9 ± 3.1	39.0 ± 2.4	38.8 ± 3.1
Preterm birth <34 weeks	9.1%	10.9%	3.8%	2.7%
Placental abruption	0.3%	1.5%	1.1%	0.8%
Preeclampsia	6.5%	5.9%	3.4%	4.2%
Diabetes	7.4%	6.5%	8.0%	9.2%
Perinatal and Childhood				
Stillbirth	12/1000	21/1000	8/1000	19/1000
Neonatal death	0	3/1000	4/1000	4/1000
NICU admission	8.6%	6.2%	11.8%	11.9%
Birthweight <10th centile	9.8%	8.1%	8.8%	7.8%
IQ median (25th, 75th percentile)	97 (85, 105)	94 (85, 107)	94 (83, 101)	91 (82, 101)

^aFor all comparisons, $p > 0.05$. EGA = estimated gestational age; IQ = intelligence quotient; NICU = neonatal intensive care unit. The printed title reads "Maternal-Fetal Medicine Unites Network" (typo for Units).

Isolated maternal hypothyroxinemia

- **Low free T4 with normal TSH; 1.3–2.1%. Low antibody prevalence (5%)** unlike subclinical hypothyroidism.
- Early reports of neurodevelopmental problems; Casey: no ↑ adverse outcomes; one large IPD analysis: ↑ preterm birth.
- **CATS and MFMU: treatment gives no benefit → ATA recommends against routine treatment.**

Euthyroid autoimmune thyroid disease

- TPO/thyroglobulin antibodies in **2–17%** of pregnant women; most are euthyroid.
- **2–5× early pregnancy loss.** Preterm birth: early RCT showed levothyroxine ↓ **22% → 7%**, but not confirmed when TSH <2.5; >1000 untreated women: no ↑ preterm birth but **3× abortion**; meta-analysis **1.3×** preterm.
- ↑ Later thyroid disease and **postpartum thyroiditis**.
- **No organization recommends universal antibody screening.**

5. Iodine Deficiency

- US population remains iodine sufficient overall, but urinary iodine in pregnant women is declining — monitor vulnerable groups.
- Needs ↑ in pregnancy: more hormone production, ↑ **renal losses**, fetal needs. Iodine is needed for fetal neurodevelopment from soon after conception; **~19 million babies/year** at risk globally.
- **Mild deficiency:** intellectual impairment doubtful, but **supplementation prevents fetal goiter**; RCT (>800 women): no effect on child development at 5–6 yr.

- Moderate deficiency: mixed data; early supplementation (200–300 µg) improved scores in some studies; >150 µg/d linked to lower psychomotor scores in another.
- **Severe deficiency → endemic cretinism**; salt iodization/supplementation clearly beneficial.
- **Recommended intake:**
 - **IOM: 220 µg/d pregnancy, 290 µg/d lactation**
 - **ATA: 250 µg/d; add 150 µg iodine to prenatal vitamins** (only half of US prenatal vitamins contain iodine)
 - **Do not exceed 500 µg/d** (excess → subclinical thyroid dysfunction, autoimmunity)

6. Congenital Hypothyroidism

- **Universal newborn screening** (since 1974, required in all states). **~1 in 2000 newborns** (rising — preterm survival, screening changes).
- **Agnesis/hypoplasia 80–90%**; the rest are hereditary hormone-synthesis defects; **up to 15% transient**.
- **Early, aggressive thyroxine** → IQ at 18 yr equal to siblings if initial dose adequate.

7. Postpartum Thyroiditis

- Transient **autoimmune** thyroiditis in **5–10%** within the first **12 months** postpartum.
- Risk rises with antibody titers: **up to 50% of women with 1st-trimester thyroid antibodies** develop it. Antibody screening still not recommended.
- Often missed (months after delivery, vague symptoms).

Phase	Timing	Features	Treatment
Thyrotoxic (destructive)	Early; abrupt; lasts a few months	Small painless goiter ; only fatigue and palpitations more common than controls	Thionamides ineffective; β-blocker if severe
Hypothyroid	Usually 4–8 months postpartum	Thyromegaly and symptoms more prominent	Levothyroxine 25–75 µg/d for 6–12 months

- Italian cohort (4562): **3.9%** developed it — **⅔ hypothyroid only, ⅓ hyperthyroid only; only 14% classic biphasic**.
- **Permanent hypothyroidism 20–30%** eventually (**3.6%/yr**); higher with high antibody titers and TSH. **Half of TPO-positive women become permanently hypothyroid by 6–7 yr**.
- Link with postpartum depression unresolved; levothyroxine didn't reduce depression in TPO-positive women.

Memory hook: "Hyper first, hypo later, most never both" — biphasic pattern only 14%; thyrotoxic phase = β-blocker, never thionamides.

8. Nodular Thyroid Disease

- Palpable nodules in up to **5%** of reproductive-age women; sonographic nodules >2 mm in 15% of Chinese gravidas (often multiple, enlarge slightly, none malignant).
- **90–95% of solitary nodules are benign**.
- Evaluate as in nonpregnant women, but **no radioiodine scanning**.
- Sonography detects nodules **>5 mm**; malignant features: **hypoechoic, irregular margins, microcalcifications**.
- **Ultrasound-guided FNA** — cytology unaffected by pregnancy. **Follicular lesion → defer surgery until postpartum**.

- **Thyroid cancer:** usually well differentiated and indolent (may persist/recur more in pregnancy). Diagnosed in 1st–2nd trimester → **thyroidectomy before the 3rd trimester**; non-aggressive or 3rd-trimester diagnosis → **defer to early puerperium**.

9. Parathyroid Disease

Calcium physiology

- **PTH** (84 amino acids) ↑ serum calcium via bone, kidney, and (indirectly) gut; regulated by ionized Ca^{2+} . **Calcitonin antagonizes PTH.**
- Fetal needs: **300 mg/d late pregnancy, 30 g total**; plus ↑ renal loss from ↑ GFR.
- **Total Ca ↓ with albumin; ionized Ca unchanged. 1,25-dihydroxyvitamin D doubles** (probably placental/decidual); maternal PTH low-normal or ↓.
- Calcium is **actively transported** across the placenta; fetal homeostasis via PTH-rP, PTH and receptors.

The extracted text calls calcitonin "a potent parathyroid hormone"; calcitonin is a thyroid (C-cell) hormone that antagonizes PTH.

Hyperparathyroidism

- **Hyperparathyroidism or cancer cause 90% of hypercalcemia.** Prevalence **2–3/1000 women** (up to 14/1000 including asymptomatic).
- **Solitary adenoma ~80%**; four-gland hyperplasia 15%; malignancy the rest (tumor PTH may not register on assays).
- Calcium usually only **1–1.5 mg/dL above normal** → only **20% symptomatic** (¼ become symptomatic as Ca rises).
- **Hypercalcemic crisis:** stupor, nausea, vomiting, weakness, fatigue, dehydration.
- **Parathyroidectomy indications (nonpregnant):** symptomatic; $\text{Ca} \geq 1.0 \text{ mg/dL above upper normal}$; $\text{CrCl} < 60 \text{ mL/min}$; ↓ bone density; age **>50**. Otherwise annual Ca/creatinine and bone density every 1–2 yr.

Hyperparathyroidism in pregnancy

- <200 reported cases; **adenoma most common**; rare ectopic PTH or carcinoma.
- Symptoms: **hyperemesis**, weakness, **renal calculi**, psychiatric disorders; sometimes **pancreatitis** presents it.
- **Pregnancy "protects"** (calcium shunted to fetus, ↑ renal excretion) → ⚠ **postpartum hypercalcemic crisis** ($\text{Ca} > 14 \text{ mg/dL}$: nausea, vomiting, tremor, dehydration, altered mental status).
- Fetal: miscarriage, FGR, LBW; historically stillbirth/preterm; **preeclampsia 30%** in one series.
- **Management:**
 - **Surgery preferred for symptomatic adenoma** — prevents fetal/neonatal morbidity and postpartum crisis. Medical care → **preterm 66%** vs **all term** after surgery (Australian series). Neck exploration tolerated **even in the 3rd trimester**.
 - Asymptomatic mild hypercalcemia → medical management; watch for postpartum crisis. Options: **calcitonin, oral phosphate 1–1.5 g/d**.
 - **Hypercalcemic crisis: IV normal saline (urine >150 mL/h) + furosemide**; prevent hypokalemia and hypomagnesemia; adjunct mithramycin.
- **Neonatal effects:** fetal hypercalcemia suppresses fetal parathyroids → **15–25% of newborns get severe hypocalcemia ± tetany** (transient; treat with calcium + calcitriol — ineffective in preterm infants). **Neonatal tetany/seizures** → **evaluate the mother for hyperparathyroidism**.

Hypoparathyroidism

- **Most common cause of hypocalcemia; ~80% follow anterior neck surgery.**

- Facial spasm, cramps, perioral/digital paresthesias → tetany, seizures. Fetal skeletal demineralization → neonatal fractures.
- **Treatment: calcitriol**, dihydrotachysterol, or vitamin D **50,000–150,000 U/d**; **calcium gluconate or lactate 3–5 g/d**; **low-phosphate diet**.
- **Goal: corrected calcium in the low-normal range** (requirements may rise or fall).

Pregnancy- and lactation-associated osteoporosis

- Average **BMD ↓ 3–4%** in pregnancy; more with breastfeeding, twins, low BMI. Parity/breastfeeding **not** linked to postmenopausal hip fracture.
- Rare disorder: **back pain late in pregnancy or postpartum**, hip pain, difficulty weight bearing.
- No cause in **>1/2**; known causes: **unfractionated heparin**, bed rest, corticosteroids; occasionally hyperparathyroidism or thyrotoxicosis.
- **Treatment:** calcium, vitamin D, analgesia; some advise stopping breastfeeding. Chronic osteopenia may persist.

10. Adrenal Gland Disorders

Pheochromocytoma

- Catecholamine-secreting chromaffin tumors, usually adrenal medulla. **"10% tumor": 10% bilateral, 10% extraadrenal, 10% malignant.**
- Associations: **MEN syndromes** (medullary thyroid carcinoma, hyperparathyroidism), **neurofibromatosis, von Hippel-Lindau.**
- **~1 in 50,000 pregnancies; 0.2–0.6%** of hypertensive patients.
- **Symptoms (paroxysmal):** hypertensive crisis, seizures, anxiety attacks, **headache, sweating, palpitations**, chest pain, nausea, vomiting, pallor or flushing. **Sustained hypertension in 60%** (half of these also have paroxysms).
- **Diagnosis:**
 - Screening: **24-h urine metanephrines and catecholamine metabolites**; diagnosis needs ≥ 2 of 3 (free catecholamines, metanephrines, VMA).
 - **Plasma catecholamine metabolites = most sensitive; normal levels exclude it.**
 - **MR imaging is preferred for localization in pregnancy.**
 - **⚠ Must differentiate from preeclampsia.**
- **Outcome:** old series **43 of 89 mothers died**; mortality **58% if undiagnosed antepartum vs 18%** if diagnosed. **Antepartum diagnosis is the key determinant of maternal survival.**

Table 61-5. Outcomes of Pregnancies Complicated by Pheochromocytoma and Reported in Four Contiguous Epochs

Factor (incidence %)	1980–87 Harper (1989), n = 48	1988–97 Ahlawat (1999), n = 42	1998–2008 Sarathi (2010), n = 60	2000–2011 Biggar (2013), n = 78
Diagnosis				
Antepartum	51	83	70	73
Postpartum	36	14	23	28
Autopsy	12	2	7	—
Maternal death	16	4	12	8
Fetal wastage	26	11	17	17

The Biggar epoch is printed as "2000–2001"; 2000–2011 appears intended given the "contiguous epochs" design.

• **Treatment:**

1. **α -blockade first: phenoxybenzamine 10–30 mg, 2–4 times daily**
 2. **Then β -blocker** (propranolol) for tachycardia — never before α -blockade
 3. **Surgery during pregnancy, preferably 2nd trimester; laparoscopic adrenalectomy** is the norm. Bilateral tumors: remove the largest during pregnancy, the other after delivery (avoids adrenal insufficiency).
 4. Diagnosed late → **planned cesarean with tumor excision** or postpartum resection.
- Recurrent tumors can cause dangerous peripartum hypertension even when controlled.



Figure 61-5. Coronal MR at 32 wk: right adrenal pheochromocytoma (arrow) beneath the liver (L) — MR is the preferred localization method in pregnancy.

Pearl: giving a β -blocker before α -blockade in pheochromocytoma leaves α -mediated vasoconstriction unopposed and can precipitate a hypertensive crisis.

Cushing syndrome

- Rare; **F:M = 3:1**. **Most cases iatrogenic** (long-term steroids).
- **Cushing disease** = ACTH-producing **pituitary adenoma** (mostly microadenomas <1 cm; half <5 mm) → bilateral adrenal hyperplasia. Ectopic CRF/ACTH from nonendocrine tumors.
- Features: moon facies, buffalo hump, truncal obesity; **fatigue/weakness, hypertension, hirsutism, amenorrhea in 75–85%**; bruising, striae; **impaired glucose tolerance up to 60%**.
- **Diagnosis:** nonsuppressible cortisol on dexamethasone, or **↑ 24-h urinary free cortisol (3 collections)**; ACTH + CT/MR to localize. **In pregnancy, use 24-h urinary free cortisol** (allowing for the normal pregnancy rise).
- Pregnancy is rare (androgen excess → anovulation). **In pregnancy, >½ are ACTH-independent adrenal adenomas**; ~30% pituitary; 10% adrenal carcinoma.

Table 61-6. Complications in Pregnancies of Women with Active and Cured Cushing Syndrome

Complication (incidence %)	Active Disease (n = 214)	Cured (n = 49)
Maternal		
Diabetes	37	2
Gestational hypertension	40	2
Preeclampsia	26	2
Cesarean delivery	52	22
Mortality	1	0
Perinatal		
Preterm birth	66	3
Fetal-growth restriction	15	5
Spontaneous abortion	11	6
Stillbirth	6	2
Neonatal death	5	0

From Caimari, 2017 (systematic review of 263 pregnancies in 220 women).

- **Treatment:** definitive = resection of pituitary/adrenal adenoma or bilateral adrenalectomy. In pregnancy:
 - Mild cases: control hypertension until delivery.
 - **Metyrapone** = interim medical therapy until postpartum surgery. **Ketoconazole** worrisome with a male fetus (blocks testicular steroidogenesis). **Mifepristone is contraindicated.**
 - **Transsphenoidal resection or laparoscopic adrenalectomy in the 2nd trimester = first line;** unilateral adrenalectomy done safely in early 3rd trimester.

Adrenal insufficiency (Addison disease)

- Needs **>90% gland destruction**. **Autoimmune adrenalitis** most common in developed countries; **TB** in resource-poor settings. **1 in 10,000–20,000 births.**
- Polyglandular autoimmune associations: Hashimoto, primary ovarian insufficiency, type 1 DM, Graves, pernicious anemia, vitiligo, alopecia, celiac sprue, myasthenia gravis.
- Untreated → infertility; weakness, nausea, vomiting, weight loss.
- **Diagnosis:** low cortisol + high ACTH. In pregnancy, **short ACTH stimulation test with <2-fold cortisol rise confirms** it.
- Diagnosis within 3 yr of delivery → ↑ preterm birth, LBW, cesarean.
- **Management:**
 - Continue glucocorticoid + mineralocorticoid replacement; **↑ dose 20–40% after midpregnancy.**
 - **Stress dose for labor, delivery, postpartum or surgery: hydrocortisone 100 mg IV q8h × 48 h.**
 - Recognize shock from other causes (hemorrhage, sepsis) promptly.

Primary aldosteronism

- **Adrenal adenoma (Conn syndrome) ~75%;** rest bilateral hyperplasia; rarely carcinoma.
- **Hypertension, hypokalemia, muscle weakness;** high aldosterone confirms.
- Pregnancy: progesterone blocks aldosterone → very high normal levels → diagnosis hard. **Renin is suppressed** with hyperaldosteronism → **aldosterone:renin ratio** helps.
- Hypertension worsens with gestation → **K⁺ supplements + antihypertensives.** Spironolactone works but has **fetal antiandrogenic** potential → **β-blockers or calcium-channel blockers preferred;** amiloride and eplerenone reported.
- **Laparoscopic resection curative;** optimal timing in pregnancy unclear.

11. Pituitary Disorders

Prolactinomas

- Pituitary enlarges in pregnancy (**estrogen-driven lactotroph hyperplasia**). **Prolactinomas = 75% of pituitary adenomas in women.**
- Normal nonpregnant prolactin **<25 ng/mL**. Features: **amenorrhea, galactorrhea, hyperprolactinemia.**
- **Microadenoma ≤10 mm; macroadenoma >10 mm.**
- Treatment: **bromocriptine** (dopamine agonist) restores ovulation; **suprasellar macroadenoma → surgery before pregnancy.**
- **Symptomatic enlargement in pregnancy: microadenoma 2.4% vs macroadenoma ~21%** (~20% in another series).
- **Surveillance:**
 - Micro: ask about **headache and visual symptoms** regularly.
 - Macro: **visual field testing each trimester.**
 - **CT/MR only if symptoms. Serial prolactin levels are useless** (normal rise).
- **Symptomatic enlargement → dopamine agonist immediately** (bromocriptine safety well established; cabergoline generally considered safe). **Surgery if no response or pituitary apoplexy.**
- Nonfunctioning adenomas can also enlarge.

The extracted text gives prolactin units as pg/mL (should be ng/mL) and says enlargement is treated with a "dopamine antagonist"; bromocriptine and cabergoline are dopamine agonists.



Figure 61-6. Sagittal T1 MR of a pituitary adenoma with a fluid–fluid level (arrow) = intratumoral hemorrhage (apoplexy).

Acromegaly

- Excess GH from an acidophilic or chromophobic adenoma. Suggested by **↑ IGF-1** — reliable mainly in the **first half** (IGF-1 rises after midpregnancy); definitive diagnosis may wait until after delivery.
- Pregnancy rare (**half are hyperprolactinemic and anovulatory**); <100 cases.
- Marginally **↑ GDM and hypertension**. Monitor for tumor enlargement; dopamine agonists less effective; **transsphenoidal resection** for symptomatic enlargement; octreotide and pegvisomant have been used.

Diabetes insipidus

- True central DI (neurohypophysis destruction/agenesis) is rare in pregnancy.

Type	Mechanism
Primary polydipsia	Suppressed vasopressin secretion
Gestational DI	↑ Degradation (placental vasopressinase) — 2–4 per 100,000 pregnancies
Nephrogenic DI	Renal insensitivity to vasopressin

- **Transient DI with AFLP** (hepatic dysfunction ↓ vasopressinase clearance).
- **Treatment: intranasal desmopressin (DDAVP)** — safe; **higher doses needed** in pregnancy (placental vasopressinase ↑ clearance).

Sheehan syndrome

- **Pituitary ischemia/necrosis after obstetric hemorrhage** → hypopituitarism; declining with modern shock care.
- **Persistent hypotension, tachycardia, hypoglycemia, lactation failure**; deficiencies may evolve — **mean time to diagnosis 13 years**.
- **⚠ Adrenal insufficiency is the most life-threatening** → assess adrenal function **immediately**; replace glucocorticoids first, then thyroid, gonadal and GH.
- Later pregnancies usually result in a live birth.

Lymphocytic hypophysitis

- Rare autoimmune lymphocyte/plasma-cell infiltration and destruction of the pituitary; **>½ of cases in women arise in pregnancy or the puerperium**.
- Hypopituitarism and/or mass effect (headache, visual field defects); sellar mass on imaging.
- **Prolactin modestly ↑, usually <100** (vs >200 with prolactinoma).
- **~30% have other autoimmune disease** (Hashimoto, Addison, type 1 DM, pernicious anemia).
- **Treatment: glucocorticoids + hormone replacement**; may be self-limited, but **>¾ need long-term replacement**.

Quick-Recall: Thyroid Test Patterns in Pregnancy

Condition	TSH	Free T4	Treat in pregnancy?
Normal 1st trimester	↓ (hCG)	High-normal	—
Overt hyperthyroidism (Graves)	↓↓	↑	Yes — PTU (to 16 wk) / methimazole
Gestational transient thyrotoxicosis / hyperemesis	↓	↑	No — resolves by midpregnancy
Subclinical hyperthyroidism	↓	Normal	No
Overt hypothyroidism	↑	↓	Yes — levothyroxine 1–2 µg/kg/d, TSH ~2.5
Subclinical hypothyroidism	↑	Normal	No benefit shown (MFMU, CATS)
Isolated hypothyroxinemia	Normal	↓	No (ATA)
Euthyroid TPO-antibody positive	Normal	Normal	No; watch for postpartum thyroiditis
Postpartum thyroiditis, thyrotoxic phase	↓	↑	β-blocker only
Postpartum thyroiditis, hypothyroid phase	↑	↓/normal	Levothyroxine 25–75 µg × 6–12 mo

Key Numbers to Memorize

- Maternal T4 = **30%** of fetal T4 at term; fetal thyroid active **>12 wk**; anti-TPO **5–15%** of gravidas
- Hyperthyroidism **0.4–1.7%**; PTU to **16 wk**; PTU:methimazole **20:1**; free T4 every **4–6 wk**
- Leukopenia **10%** (continue), agranulocytosis **0.3%** (stop), PTU hepatotoxicity **0.1%**, rash **3–5%**

- Uncontrolled thyrotoxicosis: heart failure **8%**, preterm **32%**, stillbirth **18%**; neonatal hyperthyroidism $\leq 1\%$; TRAb **>3x** normal predicts it; retest TRAb **18–22 wk**
- No pregnancy for **6 months** after ^{131}I ; ^{131}I **3 months** after stopping breastfeeding
- Thyroid storm: PTU **1000 mg** load then **200 mg q6h**; iodide **1–2 h** later; dexamethasone **2 mg q6h x 4**; lithium **300 mg q6h**
- Transient thyrotoxicosis **2–15%**; molar $\uparrow$ T4 **25–65%**; subclinical hyperthyroidism **1.77%**
- Overt hypothyroidism **0.2–1.2%**; levothyroxine **1–2 $\mu\text{g/kg}$ ($\sim 100 \mu\text{g}$)**; adjust **25–50 μg** ; target TSH **~ 2.5** ; $\frac{1}{3}$ need more from **5 wk**
- Subclinical hypothyroidism **2.3%**; isolated hypothyroxinemia **1.3–2.1%**
- Iodine: **220 μg** pregnancy, **290 μg** lactation (IOM); ATA **250 μg** , **150 μg** in vitamins, max **500 μg**
- Congenital hypothyroidism **1 in 2000**; agenesis/hypoplasia **80–90%**
- Postpartum thyroiditis **5–10%**; hypothyroid phase **4–8 months**; biphasic only **14%**; permanent hypothyroidism **20–30%**
- Solitary nodules benign **90–95%**
- Fetal calcium **30 g** (300 mg/d late); hyperparathyroid adenoma **80%**; crisis Ca **>14 mg/dL**; neonatal hypocalcemia **15–25%**; saline to urine **>150 mL/h**
- Pheochromocytoma **1 in 50,000**; "10%" tumor; mortality **58% vs 18%** (undiagnosed vs diagnosed); phenoxybenzamine **10–30 mg 2–4x/d**
- Cushing in pregnancy: adrenal adenoma **>50%**; active disease preterm **66%**
- Addison **1 in 10,000–20,000**; dose $\uparrow$ **20–40%**; stress hydrocortisone **100 mg IV q8h x 48 h**
- Conn adenoma **75%**
- Prolactinoma: micro $\leq 10 \text{ mm}$; enlargement **2.4%** (micro) vs **$\sim 21\%$** (macro); hypophysitis prolactin **<100** vs prolactinoma **>200**
- Gestational DI **2–4 per 100,000**; Sheehan diagnosis delay **13 years**